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Radiation Characteristics from The Slots Etched in X-Band Rectangular Waveguide

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Abstract. A rectangular waveguide operating in X-band (8-12GHz) is designed for the dominant mode propagation (TE_{10}) and simulated in High Frequency Simulation Software (HFSS) for verification of its dispersion characteristics and S-parameters. Annular ring slot is then etched at the longitudinal axis of the broad wall of waveguide and radiation from it is investigated in terms of gain (in dB) of antenna formed by annular ring slot. With decrease in inner and outer radius of annular slot, the gain decreases because smaller slot disturbs surface current to a lesser extent. Radiation decreases with offset of slot from its longitudinal axis. Two similar annular ring slots are etched along the longitudinal axis with separation of quarter wavelength and half wavelength forming an array resulting in broadside and endfire radiation respectively. Effect on the nature of radiation pattern with increase in the number of annular ring slots due to non-uniform excitation is studied. However, impedance matching is not considered in this work.

Keywords: Annular ring slot, broadside, surface current, X-band waveguide.

1 Introduction

Transmission lines are used for carrying electrical and electromagnetic signals from one place to another. The most common type of transmission line is twin wire line used at low frequencies (upto 300MHz). At higher frequencies, the loss from the conductor increases and also there is unwanted radiation from these conductors. Coaxial wires are used at higher frequencies as the outer shielded conductor prevents radiation from inner conductor from leaking out of the line. A better solution is to have a single conductor, instead of two, to prevent loss from conductor [1]-[4]. A waveguide is one such transmission line which is most commonly used at frequencies above 3GHz because at frequencies lower than 3GHz, the size of waveguide becomes too heavy and bulky to handle and to integrate it with other systems especially when size and weight of the whole system matters. A waveguide offers very low loss of signal. A design of waveguide should ensure waveguide supports only one mode in the desired frequency range. This can be ensured by choosing proper cross sectional

dimension of waveguide. Rectangular waveguides are most commonly used because of its simple design.

Etching slots in waveguide leads to radiation from the slots. This principle can be used to design slotted waveguide antennas. Such antennas are commonly used in those areas which require high power as waveguides have high power handling capability and high efficiency. Some applications include satellite communication, radar systems, navigation systems, remote sensing etc. However, impedance matching is not considered in this work.

2 Literature Survey

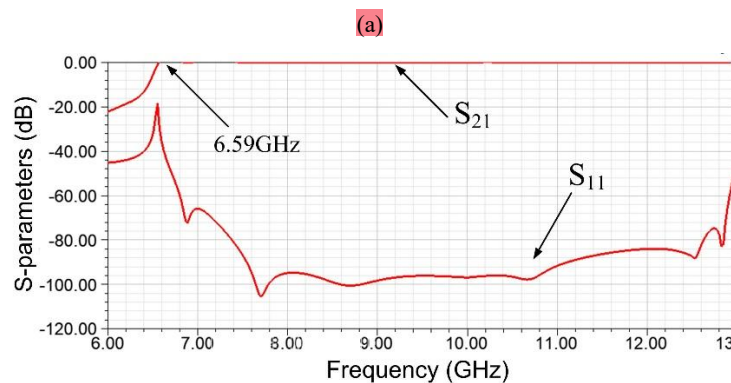
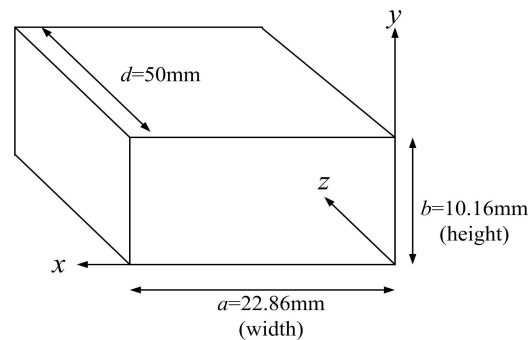
Many researchers in the past have reported slotted waveguide antenna considering rectangular slots. The authors in [4] have proposed a textile based slotted waveguide antenna array operating in 5.8GHz ISM band. Three rectangular slots have been cut on the broad waveguide walls at an offset from longitudinal axis and their dimensions and positions optimized for impedance matching. An impedance bandwidth of 300MHz has been obtained at 5.8GHz. The waveguide walls are made of conductive fabric. This antenna can be easily worn with jackets, belts etc. The authors in [5] have presented similar slotted waveguide antenna in the frequency range 14-14.5GHz in order to couple power to plasma cavity. Here also arrays of rectangular slots have been etched in the broad waveguide walls. Traveling-wave slotted waveguide array working in the frequency range 2-2.45GHz and 12GHz have been reported by authors in [6]. Here 9-element rectangular slits have been cut in the broad waveguide walls. The authors in [7] have proposed 8 rectangular slots with round edges (instead of sharp edges) in the conducting wall of the rectangular waveguide antenna. The slots are positioned in such a way so as to provide Chebychev distribution in order to provide low side-lobe level of around -30dB. The operating frequency range of the antenna is from 33.6GHz to 35.9GHz with the maximum gain of 13.4dBi. The authors in [8] have reported S-band slotted waveguide antenna for high power microwave applications. 10 slots have been etched on the broad walls of waveguide. Two inside grooves are also cut in waveguide walls. The slots are positioned and oriented in such a way so as to form Gaussian distribution of radiated fields from these slots. Side-lobe level of -20dB in H-plane and -22dB in E-plane is obtained. The designed antenna radiates at 3GHz. The authors in [9] have proposed slotted-waveguide antenna array in L band where the slots are encapsulated in a dielectric which minimizes electric field magnitude in air thus providing high power handling capability. The authors in [10] have used Chebyshev, nonomial and Taylor distribution to design a broadwall slotted waveguide antenna with the desired Side Lobe Level and the operating frequency. The procedure is able to calculate the slot length, width and location along the length of waveguide and offset from the central axis of the guide.

However, none of the authors have demonstrated radiation from annular ring slot in the waveguide. The authors feel that annular ring might go a long way as an important structure since two such concentric rings with splits leads to split ring resonator and complementary split ring resonator which finds numerous applications

in improving the characteristics of several microwave components. So the current work aims to provide some preliminary investigation into the radiation from annular ring slots in waveguides.

3 Annular Slots Design and Results & Discussion

A rectangular waveguide is designed at X-band covering frequency from 8 to 12GHz. Waveguide is considered air-filled, hence no dielectric is used, and air is considered as dielectric. Considering TE_{10} mode propagation as the dominant mode and using well known expressions of cutoff frequency of air-filled rectangular waveguide [2], the dimensions of rectangular waveguide are 22.86mm (broad wall dimension or width of waveguide) and 10.16mm (narrow wall dimension or height of waveguide). The rectangular waveguide is designed in HFSS and its S-parameters, E-field distribution, surface current at 10GHz for the dominant TE_{10} mode has been studied as shown in Figure 1. From the dispersion plot shown in Figure 1(c), it is clear that the cutoff frequency of waveguide for TE_{10} mode is 6.59GHz whereas for next higher mode (TE_{01} mode) it is 13.1GHz. Thus it is guaranteed that only one dominant mode propagates in X-band. The surface current distribution (vector form) and magnitude of E-field distribution is shown in Figure 1(d) and 1(e) respectively.



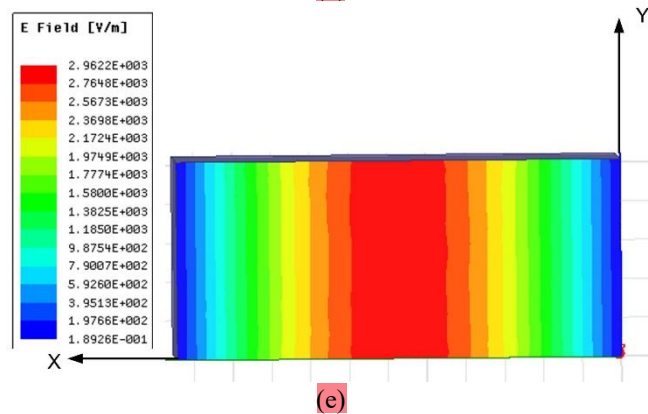
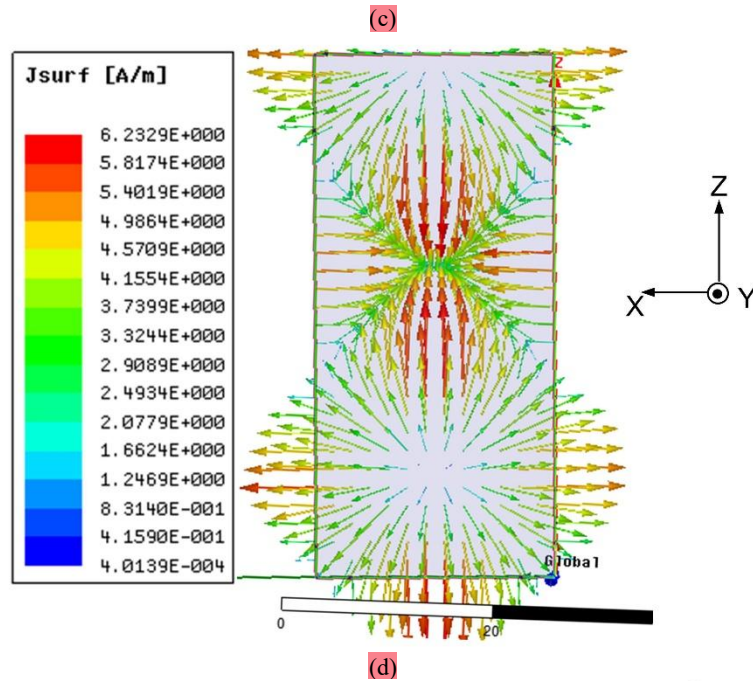
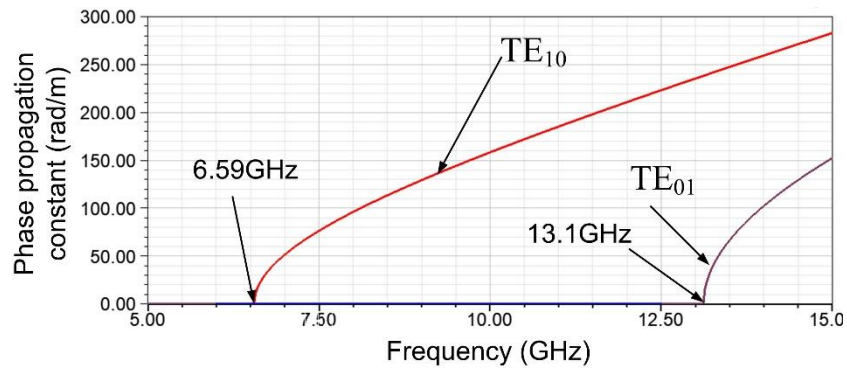
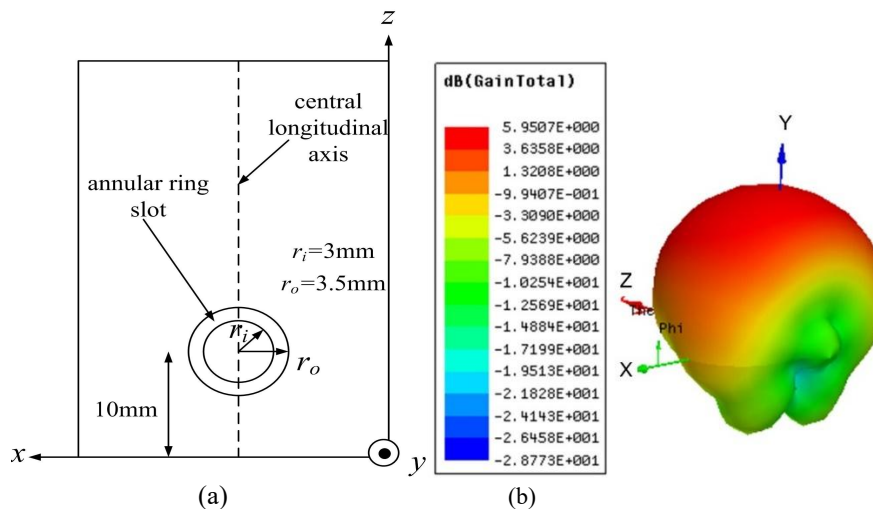


Fig. 1. (a) Dimensions of rectangular waveguide, (b) S-parameter plot for the dominant TE₁₀ mode, (c) Dispersion curve for the dominant mode (TE₁₀) and the next higher mode (TE₀₁), (d) surface current (vector from) along waveguide wall at 10GHz for the dominant TE₁₀ mode, and (e) magnitude of E-field distribution at 10GHz for the dominant TE₁₀ mode.

Now an annular ring of thickness 0.5mm is etched on the broad wall side of the rectangular waveguide along longitudinal axis as shown in Figure 2(a). Let the inner radius of the annular be 3mm (denoted by r_i) and the outer radius be 3.5mm (denoted by r_o). It is observed that radiation occurs from this slot, as is evident from the 3D radiation pattern of the antenna formed due to this slot as shown in Figure 2(b). Radiation occurs due disturbance of the surface current as shown in Figure 2(c). With increase/decrease in the size of the annular ring slot the radiation from it increases/decreases as is evident from the gain of the antenna shown in Table 1. The reason for negative gain when the size of the annular ring is very small is that the annular ring slot is unable to disturb the surface current. When the annular ring is given some lateral offset from longitudinal axis (along x-axis), the radiation from the slot decreases because E-field intensity is maximum at the centre and decreases away from the centre for TE₁₀ mode. However, there is no effect of changing the annular ring slot longitudinally as far as radiation pattern is concerned because the wave is traveling along the waveguide. But impedance may be affected. This work does not take into account the impedance matching at the source end of the waveguide.



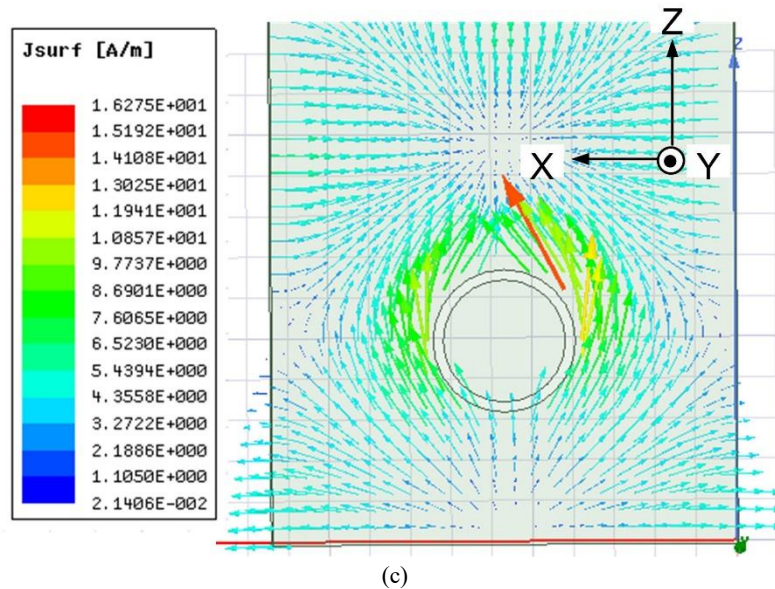


Fig. 2. (a) Annular ring slot etched on the conductor on the broad wall side of waveguide at the longitudinal axis of waveguide, and (b) Simulated 3D radiation pattern of the antenna formed due to radiation from the annular slot, and (c) surface current around the annular slot at an instant of time at 10GHz for the dominant TE_{10} mode.

Table 1. Data showing the variation of gain of antenna with the size of the annular ring keeping the thickness of the annular slot constant at 0.5mm and distance from one end of the waveguide as 10mm

Outer radius of the annular ring (r_o) (in mm)	Inner radius of the annular ring (r_i) (in mm)	Gain (dB)
2.5	2	-1.77
3	2.5	1.715
3.5	3	4.5
4	3.5	5.897
4.5	4	6.5
5	4.5	6.6
5.5	5	7.2

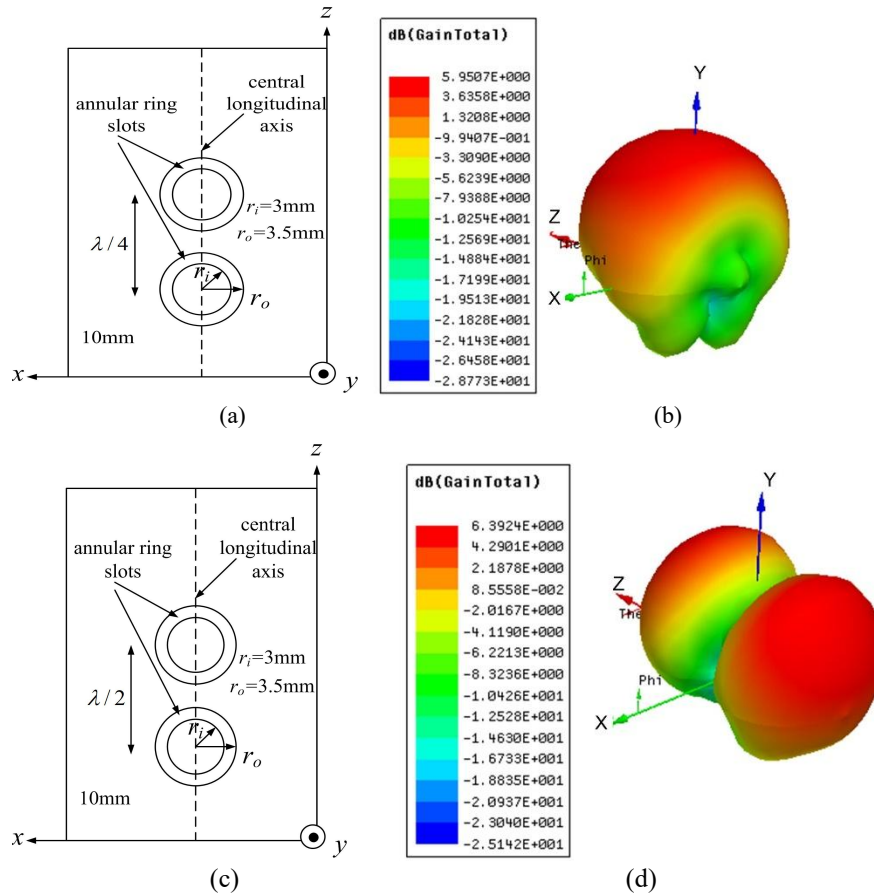
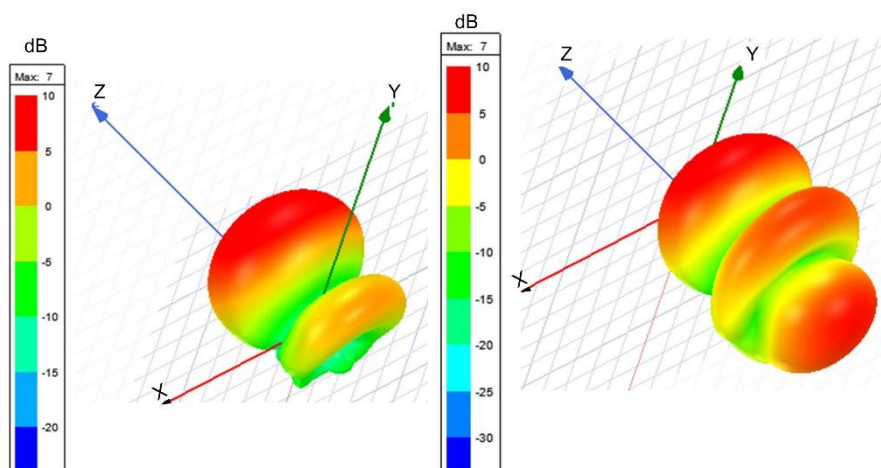
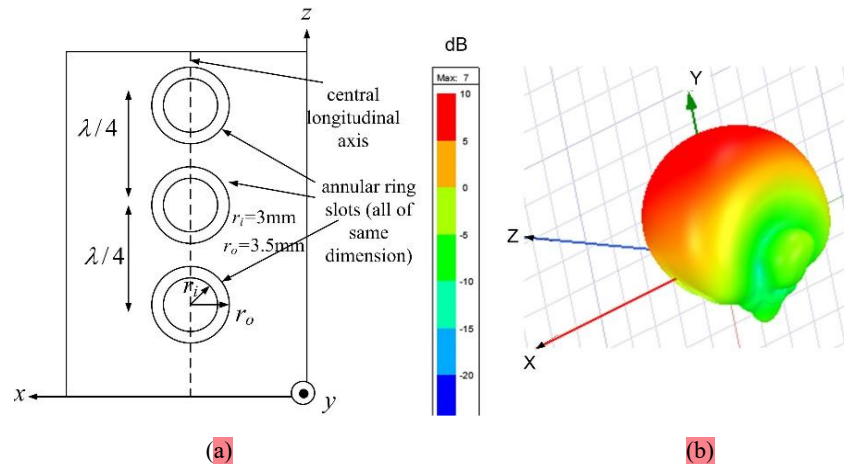


Fig. 3. (a) Two annular ring slots separated by distance $\lambda/4$ (λ is the free space wavelength), (b) Simulated 3D radiation pattern of two slots shown in (a), (c) Two annular ring slots separated by distance $\lambda/2$, and (d) Simulated 3D radiation pattern of two slots shown in (c) at 10GHz.

The effect of two annular slots is considered on the radiation pattern. Two cases have been considered. In the first case two annular slots are separated by a distance of quarter free-space wavelength (which is around 7.5mm) as shown in Figure 3(a). Broadside radiation is obtained with a maximum gain of 5.95dB as shown in Figure 3(b). In another case, the same two annular slots are now separated by half free-space wavelength (which is around 15mm) as shown in Figure 3(c). Here endfire radiation is obtained with a maximum gain of 6.39dB as shown in Figure 3(d).

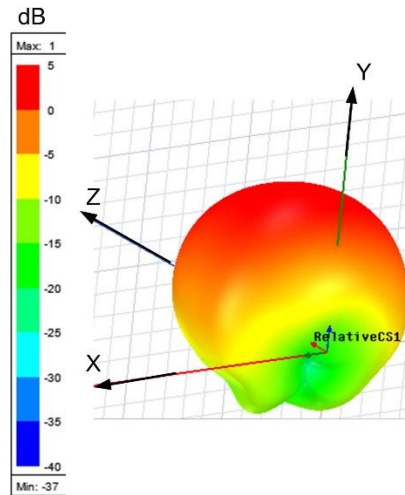
However, if three such annular ring slots are etched longitudinally and separated by quarter-wavelength (Figure 4(a)), then end-fire radiation pattern is achieved as shown in Figure 4(b) with maximum radiation along z-direction. The reason for the change of radiation pattern from broadside to end-fire with increase in the number of slots, though keeping the distance between the slots same, is due to the fact that the

magnitude of the electric field (hence also magnetic field) changes along the waveguide axis, leading to unequal excitation of all ring slots. When the distance the annular rings is increased to 0.4λ , the radiation is still endfire but a minor lobe starts to develop in the direction opposite to maximum radiation as shown in Figure 4(c). Now when the separation between the annular rings is increased to 0.5λ , multiple lobes are obtained with maximum radiation both along waveguide axis and in broadside direction as shown in Figure 4(d). However when the distance between the annular rings is less than a quarter wavelength (approx. 0.18λ), broadside radiation pattern is achieved as shown in Fig. 4(e). Finally, three annular slots are etched in the transverse direction as shown in Figure 5(a) with quarter-wavelength separation between the slots. Here broadside radiation is observed as shown in Figure 5(b). Here also, due to TE_{10} mode excitation, the magnitude of electric-field (hence also magnetic field) excitation for the annular ring slots varies. So the excitation is non-uniform for all the annular ring slots.



(c)

(d)



(e)

Fig. 4. (a) Three annular ring slots separated by distance $\lambda/4$ (λ is the free space wavelength which is 3cm), (b) Simulated 3D radiation pattern of three slots shown in (a) with maximum gain 7dB at 10GHz, (c) Simulated 3D radiation pattern of three slots when separated by 0.4λ with maximum gain 7dB at 10GHz, and (d) Simulated 3D radiation pattern of three slots when separated by 0.5λ with maximum gain 7dB at 10GHz (e) Simulated 3D radiation pattern of three slots when separated by 0.18λ with maximum gain 1dB at 10GHz

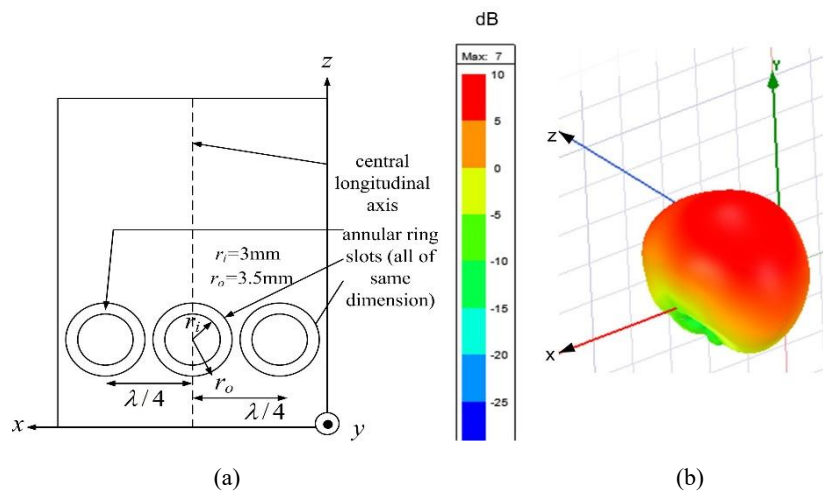


Fig. 5. (a) Three annular ring slots arranged transverse to waveguide axis separated by distance $\lambda/4$ (λ is the free space wavelength which is 3cm), (b) Simulated 3D radiation pattern of three slots shown in (a) with maximum gain 7dB at 10GHz

The analysis on antenna array available in texts consider antennas to be point elements. So the results may deviate if the geometry from which radiation occurs has some dimension. Here, the distance between elements is comparable to the dimension of annular slots. Moreover, the phase and amplitude excitation to each of the annular rings both change along the waveguide, so the prediction of broadside and endfire radiation requires a deeper analysis of the structure which has been kept outside the purview of this work. The structure reported here comes under the class of traveling-wave antennas which requires different treatment. The work is limited to reporting important findings regarding the radiation pattern from the annular ring slots and effect of spacing between annular ring slot arrays.

4 CONCLUSION & FUTURE SCOPE

A rectangular slot etched longitudinally at the centre of the broad wall dimension of waveguide along longitudinal axis does not radiate but an annular ring slot at the same location does radiate. The amount of radiation is proportional to the size of the slot because larger size slot disturbs current distribution to a greater extent as compared to the smaller size slot. The amount of radiation from the slot also depends on the E-field intensity through the waveguide. A number of similar slots etched on the conductor of rectangular waveguide behave like an antenna array and radiation from such array may be broadside or endfire depending on the distance between the slots and the uniformity of excitation for all the annular ring slots. This work is restricted only to studying the radiation pattern from the annular ring slots. Further work may be done by providing impedance matching techniques. Also effect of annular slot arrays of varying sizes may be studied. An empirical relation between the non-uniform excitation inputs and the nature of the radiation pattern may be established after carrying out good number of simulations. Moreover, nature of radiation pattern may be analytically established and results compared with simulation.

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